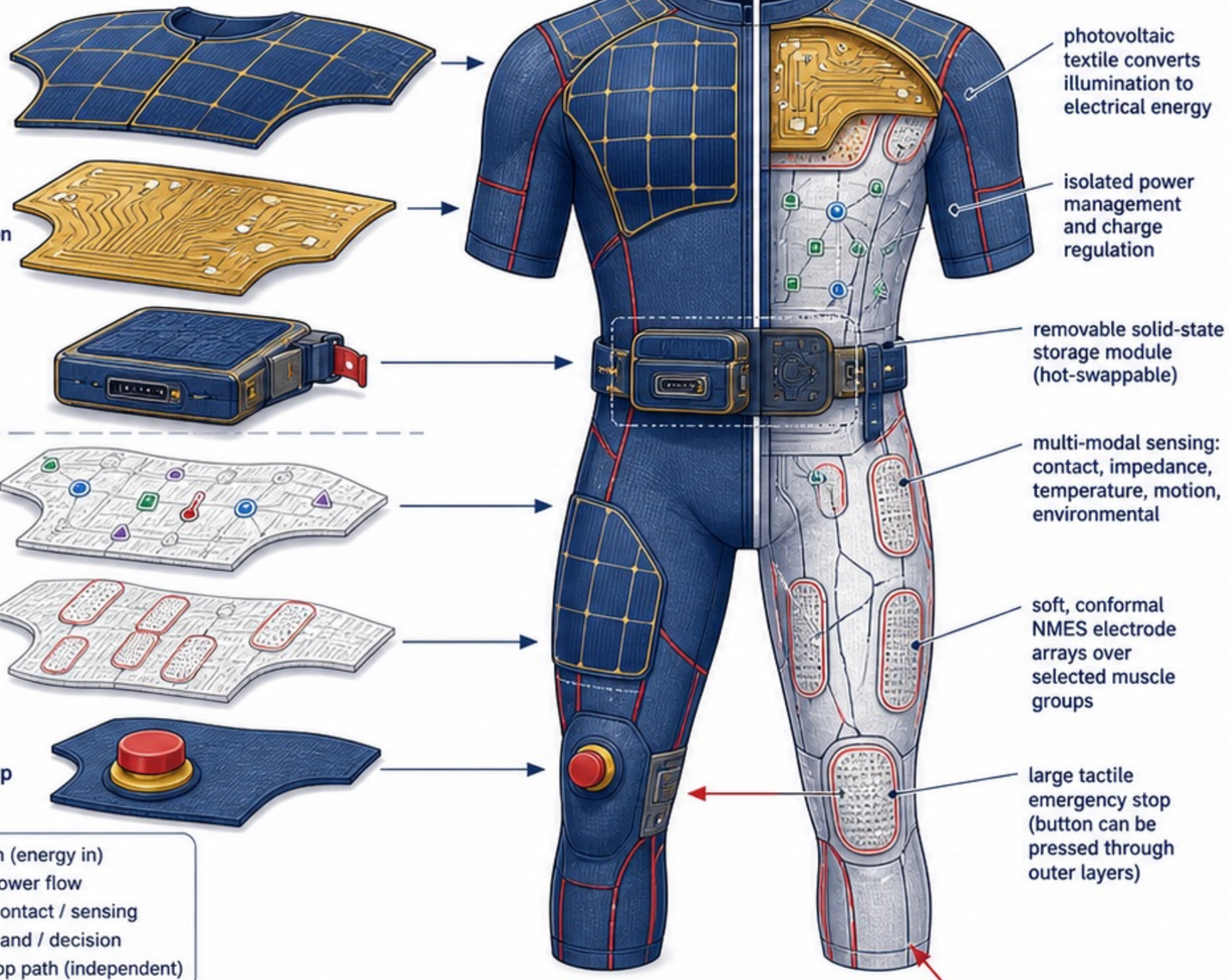
ILLUSTRATIVE SUIT DESIGN DETAIL

ENERGY LAYER

1. photovoltaic textile shell
2. isolation + charge regulation
3. removable storage

BODILY LAYER

4. contact / impedance sensing
5. soft NMES electrodes
6. local emergency stop



- photovoltaic textile converts illumination to electrical energy
- isolated power management and charge regulation
- removable solid-state storage module (hot-swappable)
- multi-modal sensing: contact, impedance, temperature, motion, environmental
- soft, conformal NMES electrode arrays over selected muscle groups
- large tactile emergency stop (button can be pressed through outer layers)

illumination (energy in)
reserve / power flow
electrode contact / sensing
user command / decision
physical stop path (independent)

CONTROLLER / INTERFACE

- **preview** — user reviews system status, energy reserve, sensor health, and environment
- **affirmative initiation** — user affirms to arm NMES support for contingency use
- **monitor** — real-time monitoring of user state, energy reserve, and system performance
- **stop / degrade** — user or system stops or degrades NMES support as needed

USER + CLINICAL LIMITS

- not flight qualified
- no autonomous background stimulation
- contingency support only

PHYSICAL STOP PATH (INDEPENDENT OF DISPLAY)

- press local emergency stop (large tactile button)
- immediate hardware interrupt to NMES driver outputs
- all stimulation outputs disabled (regardless of controller state)

DATA + ENERGY FLOWS (LOGICAL)

